CrewAI Production Security Baseline

Prepared for: [Enterprise Engineering Lead]
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Executive Summary

CrewAI has **nine CVEs in 2026**, including an **unpatched zero-day** (CVE-2026-92206, CVSS 8.8) published by Trend Micro's Zero Day Initiative after **11 months of vendor silence**. The project has **zero published GitHub Security Advisories** despite confirmed RCE, sandbox escape, and SSRF flaws.

If your organisation runs CrewAI in production — especially with Code Interpreter, RAG tools, or agent configs loaded from repositories — you are running unpatched critical vulnerabilities.

Finding 1: Unpatched Zero-Day — Unsafe Reflection RCE

Field	Detail
CVE	CVE-2026-92206
CVSS	8.8 (High)
CWE	CWE-470 — Use of Externally-Controlled Input to Select Classes or Code
ZDI Advisory	ZDI-26-706
Status	**No patched version exists.** Vendor unresponsive for 11 months.
Exploitation	Requires victim to load malicious agent config from a repository
CISA KEV	Not listed

Mechanism: `load_agent_from_repository()` does not restrict user-supplied arguments before importing a module. Loading a malicious agent configuration from a Git repo, S3 bucket, or shared drive results in remote code execution in the service account's context.

Why this matters for production: - CI/CD pipelines that pull agent configs from repos are exposed - Multi-tenant platforms where users can upload agent definitions are exposed - Any agent config path influenceable by a prompt is an attack surface

Current status as of 2026-10-31: No fixed version named. PyPI shows 1.15.22 as latest with no changelog reference to this CVE. The GHSA entry lists no patched version. Dependabot will not alert on this.

Finding 2: Code Interpreter Sandbox Escape

Field	Detail
CVE	CVE-2026-2275
CVSS	9.8 (Critical)
Mechanism	Code Interpreter falls back to unsandboxed execution when Docker is unreachable
Status	Fixed in commit `fb2323b` — Docker now mandatory unless `unsafe_mode=True`

Why this matters: If your deployment runs CrewAI without Docker, or if Docker is temporarily unavailable, the Code Interpreter executes arbitrary Python/bash as root. The fix exists but many production deployments have not upgraded.

Evidence path: Fix commit enforces Docker as mandatory. Execution raises `RuntimeError` when Docker is missing unless `unsafe_mode=True` is set. Many deployments set this flag for convenience.

Finding 3: SSRF via RAG Search Tools

Field	Detail
CVE	CVE-2026-2286
CVSS	High

Field	Detail
Mechanism	RAG search tools don’t validate runtime-provided URLs
Impact	SSRF against internal services and cloud metadata endpoints (169.254.169.254)

Why this matters: A prompt-injected URL in a RAG query can reach internal services, cloud metadata, and databases. This is the same class as CVE-2026-33626 (LMDeploy) which is actively exploited.

Finding 4: Path Traversal via JSON Loader

Field	Detail
CVE	CVE-2026-2285
CVSS	High
Mechanism	JSON loader tool reads files without path validation
Impact	Arbitrary local file read, including `/etc/passwd`, `.env` files, private keys

Why this matters: A prompt-injected file path can read any file the CrewAI process can access. This includes environment files with API keys, database credentials, and cloud tokens.

Finding 5: Silent Sandbox Downgrade

Field	Detail
CVE	CVE-2026-2287
CVSS	Medium
Mechanism	No continuous verification that Docker is running during execution
Impact	Silent downgrade to insecure sandbox mode

Why this matters: Even if Docker is running at startup, a crash or restart can leave the Code Interpreter in unsandboxed mode with no alert.

Finding 6: GitHub Security Advisory Gap

Fact: CrewAI has **zero published GitHub Security Advisories** as of 2026-10-31.

Implication: Dependabot, Renovate, and other automated dependency tools will not alert on these CVEs. The project's own security infrastructure does not track its vulnerabilities. This is a governance failure that extends to every downstream project using CrewAI.

Dependency Chain Exposure

CrewAI depends on: - **LangChain** — 13 CVEs in 90 days including CVE-2026-106119 (MongoDB injection) - **LangGraph** — 3 CVEs including SQLite/Redis injection and deserialization RCE - **Pydantic** — multiple CVEs in deserialization - **OpenAI / Anthropic SDKs** — API key leakage vectors

Standard `pip install crewai` pulls all of these. A standard deployment inherits the entire vulnerability surface.

Production Deployment Checklist

Before you deploy CrewAI

Check	Status	Action
`crewai` version		Pin to latest. Check GHSA for CVEs.
`crewai-tools` version		Pin to >= 1.15.1 (SSRF fix).
Docker required for Code Interpreter		Enforce. Never set `unsafe_mode=True`.
Agent config source		Load only from reviewed, signed repositories.

Check	Status	Action
MCP servers in config		Audit each server. Remove unapproved.
Network egress from CrewAI container		Restrict to LLM API endpoints only.
Environment variables in process scope		Use runtime injection, not `.env` files the agent can read.
Logging		Enable audit log for all tool calls with parameters.
CI/CD pipeline		Scan agent configs for untrusted sources before deployment.
Dependency scanning		Use sentinel or equivalent. CrewAI GHSA is empty — don’t rely on it.

Remediation Playbook

Immediate (24 hours)

1. **Upgrade `crewai` and `crewai-tools`** to latest versions. Confirm Docker is running wherever Code Interpreter is used.
2. **Audit all agent config sources.** Remove any loaded from repos, S3, or shared drives not under your control.
3. **Rotate all credentials** accessible to the CrewAI process.

Short-term (1 week)

1. **Implement MCP server allowlist.** Only approved servers with signed manifests.
2. **Add URL validation** to all RAG/search tools. Block private IP ranges and metadata endpoints.
3. **Add path validation** to all file-reading tools. Restrict to project directory.

Medium-term (1 month)

1. **Deploy continuous dependency monitoring.** CrewAI's GHSA is empty — use external feeds (CISA KEV, NVD, sentinel).
2. **Conduct prompt injection testing** against all CrewAI workflows.
3. **Review multi-tenant isolation** if you run CrewAI as a platform.

What Twinthos Delivers

Service	Deliverable	Price
CrewAI Production Security Baseline	Full CVE audit, dependency chain analysis, sandbox configuration review, MCP server inventory, remediation playbook	£8,000
Ongoing CrewAI Monitor	Continuous CVE tracking, config drift alerts, Telegram notifications	£500/month
Prompt Injection Test Suite	Automated canary prompts against all agents, weekly re-tests	£2,000/setup + £500/month

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Contact

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This audit was prepared using the Twinthos AI Security Sentinel, which cross-references CISA KEV, FIRST EPSS, and dependency lockfiles against real-time CVE feeds.